

5 Animals and plants can detect and respond to light.

(a) Rod cells respond to light by producing action potentials in the optic nerve.

(i) In which structure are rod cells located?

(1)

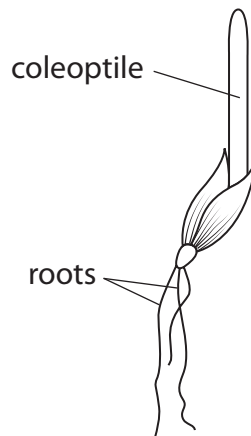
- ☐ **A** iris
- ☐ **B** pupil
- ☐ **C** retina
- ☐ **D** spinal cord

(ii) Describe the role of rhodopsin in rod cells.

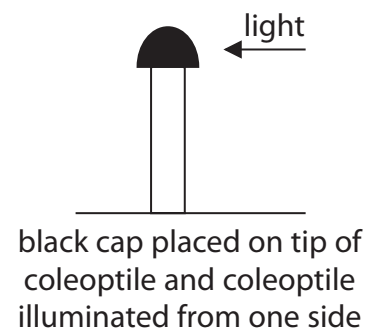
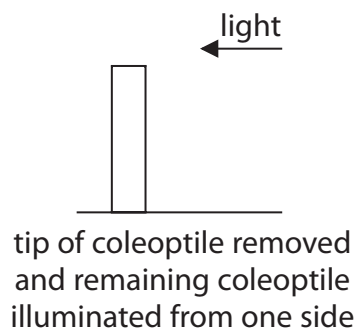
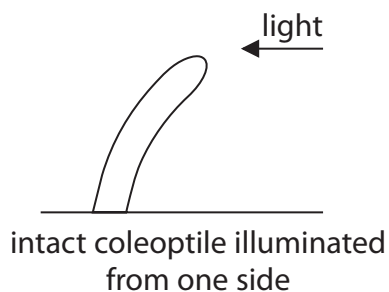
(3)

*(b) The drawing shows the results of some experiments investigating phototropisms.

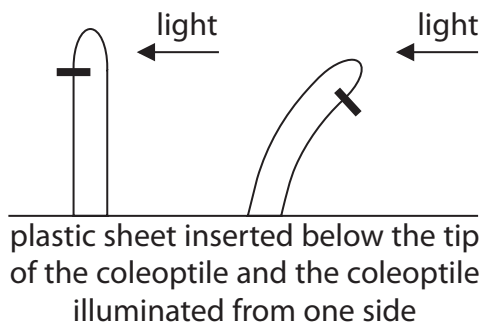
The diagram shows a coleoptile of a seedling.



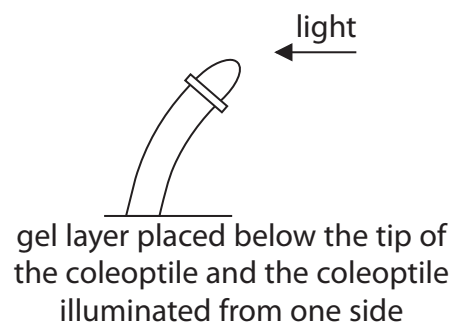
Experiment 1:



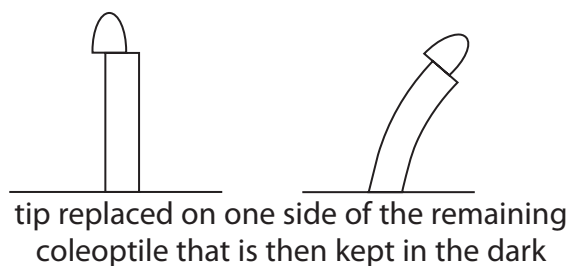
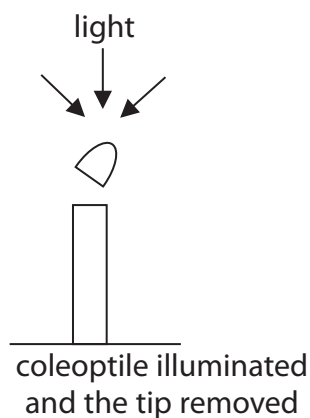
Experiment 2a:



Experiment 2b:



Experiment 3:



Scientists concluded that these results suggest a plant growth substance is involved in regulating the phototropic response.

Discuss how the results of these experiments support the role of a plant growth substance in the phototropic response of plants.

Use the information for these experiments and your own knowledge to support your answer.

(6)

[illegible]

(Total for Question 5 = 10 marks)